

# Digital Information Processing Lab 2025



Class 3: Vocoder and STFT

Group 3 : The Smarties

26.05.2024

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## ✓ Preparatory task

### Question 1

Define the Short Time Fourier Transform (STFT) and explain its purpose in signal processing. How does it differ from the regular Fourier Transform?

**Ans:**

Defination: The Short Time Fourier Transform (STFT) is a mathematical tools used to analyze the content of frequency of non-stationary signals (A statinary signal is a signal whose statistical properties like mean, Frequency content, Amplitude changes over time being ) in digital processing. This method is used to represents the signal into time and frequency domain at a time.

The STFT for a signal called  $x(t)$  defined as

$$\text{STFT}\{x(t)\}(t, \omega) = X(t, \omega) = \int_{-\infty}^{\infty} x(\tau)w(\tau - t)e^{-j\omega\tau} d\tau$$

where:

### STFT Formula Components

- $x(t)$ : the input signal
- $w(t)$ : a window function (e.g., Hamming, Rectangle) centered at time (  $t$  )
- $\omega$  (or  $f$  for frequency) is the angular frequency.
- $X(t, \omega)$  is the STFT of the signal  $x(t)$

Applications:

STFT is widely used in various field, for example

- Speech processing: Analyzing and recognizing speech pattern.
- Audio Processing: Analyzing and manipulating music signals.
- Sonar: Analyzing underwater sounds.
- Mechanical Vibration analysis: To track and identify the harmonics components in noise signals.
- Radio communication: To Displaing and analyzing the radio signals.
- Fault detection in mechanical systems.

One example of STFT is discussed below to have clear idea regarding its application.

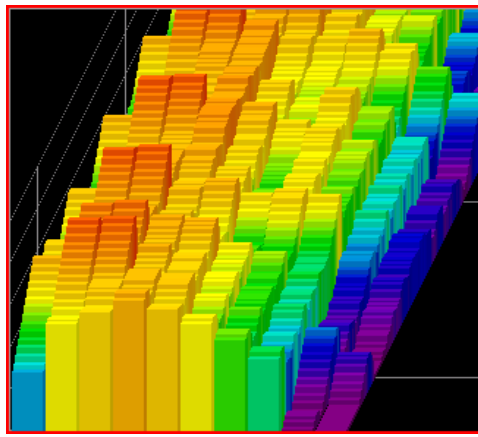


Fig. 1: An STFT being used to analyze an audio signal across time

[1]

Explanation:

Short-Time Fourier Transforms (STFTs) and other spectral tools are commonly used in music analysis. In a spectrogram, frequency is shown across the horizontal axis (low to high), amplitude is represented by bar height and color, and time appears as depth or sequential bars. Audio engineers use these visuals to identify specific frequencies, resonances, and noise components in recordings, which helps in tasks like equalization and audio effect tuning.

[1]

Difference between STFT and an ordinary Fourier Transform:

The key difference between an ordinary Fourier Transform and STFT is that in STFT method we only analyze our signal in short overlapping time segments while in other method we thoroughly examine our signal in one go. This allows us to analyze both time and frequency information, showing how frequency components get affected over time. This is not the case in Standard Fourier analysis as it only provides frequency information over the signal. FT is better for stationary signals while we can also analyze non-stationary signals with the help of STFT.

| Feature              | STFT                                              | Fourier Transform (FT)                             |
|----------------------|---------------------------------------------------|----------------------------------------------------|
| Input signal type    | <b>Non-stationary</b> (time-varying spectra)      | <b>Stationary</b> (time-invariant spectra)         |
| Time resolution      | <b>Preserved</b> (via time windows)               | <b>Lost</b> (entire signal is treated as one unit) |
| Frequency resolution | Depends on window size                            | Higher, for long signals                           |
| Output               | Function of both <b>time</b> and <b>frequency</b> | Function of <b>frequency</b> only                  |
| Common use cases     | Speech, music, transient detection                | Periodic or global spectral analysis               |

[1]

## Question 2

Discuss the concept of time-frequency analysis and explain why it is useful in analyzing non-stationary signals.

**Ans:**

Time-Frequency Analysis is a signal processing technique used to analyze signals whose frequency characteristics change over time (which is popularly known as non-stationary signals). Unlike stationary signals which have static properties like mean, variance over time which do not change with time but, in case of non-stationary signals it requires some advanced method to come up with the frequency information at different time intervals.

To sum up, we can say that we can't get the required information for non-stationary with FT method so we need Time-Frequency Analysis.

To determine this we have several tools like:

- Short-Time Fourier Transform (STFT)
- Wavelet Transform (WT)

- Wigner-Ville Distribution (WVD)
- Hilbert-Huang Transform (HHT)

### Why it's useful for Non-Stationary Signals:

- To deal with real-world signals are often non-stationary:
  - Biomedical signals (for EEG, ECG)
  - Mechanical vibrations
  - Communication signals
  - Radar and Sonar
- Captures fleeting characteristics that pure frequency-domain analysis cannot see.
- Makes time-localized spectrum analysis possible, which is essential for uses such as:
  - Fault detection in machines.
  - Speech recognition.
  - Seismic analysis.
  - Time-varying spectral estimation in communication systems.

### Question 3

What is the spectrogram and how is it related to the STFT? Explain how the spectrogram can be used to visualize time-varying frequency content in a signal.

**Ans:**

A **spectrogram** provides a visualisation of how the frequencies within a signal change over time. It's typically displayed as a graph where the abscissa axis represents time, the ordinate axis represents frequency, and the color or intensity of any point on the graph shows the strength (amplitude) of a specific frequency at a particular instant. Example of a typical spectrogram illustrated in the Figure 2 below:

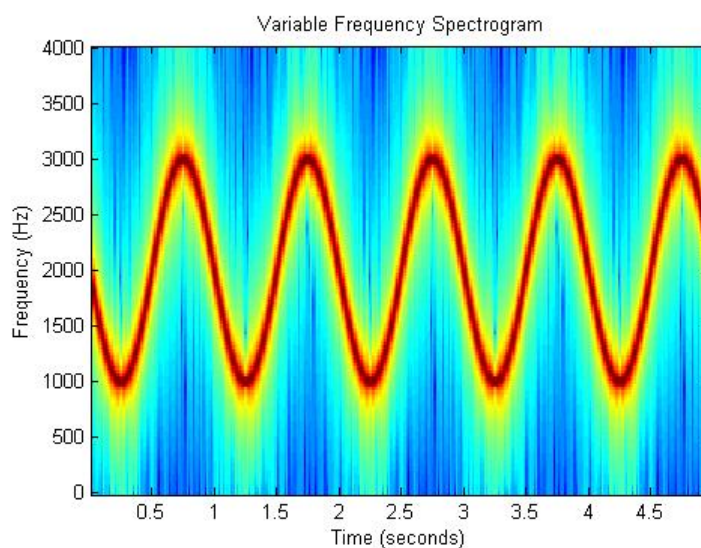


Fig. 2: Spectrogram

[\[2\]](#)

The creation of a spectrogram relies heavily on the **Short-Time Fourier Transform (STFT)**. The STFT is a technique used to determine the frequency components of localized sections of a longer signal. The relationship is as follows:

1. **Signal Segmentation (Windowing):** The STFT breaks down the main signal into smaller, often overlapping, segments. Each small segment is typically multiplied by a "window function" (like a Hann or Hamming window). This step helps to minimize spectral artifacts by smoothly tapering the edges of each segment before analysis.
2. **Frequency Analysis (FFT):** The Fast Fourier Transform (FFT), an efficient algorithm to compute the Discrete Fourier Transform, is then applied to each of these windowed segments. This transforms each short time-slice of the signal

from the time domain into the frequency domain, showing the amplitude and phase of the frequencies it contains.

3. **Spectrogram Construction:** The STFT process results in a series of frequency spectra, one for each analyzed segment. The **spectrogram is formed by arranging these individual spectra side-by-side**. Usually, it plots the magnitude squared (which represents power) or just the magnitude of the frequency components found by the STFT. Each vertical line or column in the spectrogram image corresponds to the frequency spectrum of one short segment of the signal.

[\[3\]](#)

### Visualizing Time-Varying Frequency Content:

The spectrogram is powerful because it allows us to observe the evolution of a signal's frequency makeup:

- The **time axis** (usually horizontal) lets you track when changes occur.
- The **frequency axis** (usually vertical) shows what frequencies are present.
- The **color or intensity** at any specific time-frequency point indicates how prominent that frequency is at that moment.

For example, analyzing speech with a spectrogram can reveal how resonant frequencies, known as formants, shift as different vowels are spoken. In music, it can visually distinguish melodies, harmonies, and the characteristic sounds of various instruments over a piece's duration. The choice of the segment length (window size) in the STFT is important: short segments offer precise timing of frequency events but blur distinct frequencies, while longer segments give clearer frequency detail but less precise timing. [\[3\]](#)

### Question 4

What is the Phase Vocoder and what is its role in time stretching or pitch shifting audio signals? Explain the basic principles behind the Phase Vocoder algorithm.

**Ans:**

The **Phase Vocoder (PV)** is a sophisticated audio processing algorithm that operates in the frequency domain, primarily using the Short-Time Fourier Transform (STFT). It's designed to analyze, modify, and then resynthesize an audio signal, allowing for independent adjustments to its duration and pitch. The concept was pioneered in early digital signal processing. [\[4\]](#) Its main applications are **time stretching** (making audio longer or shorter without changing its pitch) and **pitch shifting** (making the pitch higher or lower without changing its duration).

### Role in Time Stretching and Pitch Shifting:

- **Time Stretching:** To change an audio signal's length without altering its pitch, the Phase Vocoder analyzes the input sound with a specific STFT window advancement (analysis hop size). It then reconstructs the sound using a *different* window advancement (synthesis hop size). If the synthesis hop is larger than the analysis hop, the sound becomes longer; if smaller, the sound becomes shorter. A critical part of this process is the careful adjustment of phase information from one analyzed segment to the next during reconstruction. This phase coherence is vital to prevent audible artifacts, an area where significant improvements have been made. [\[3\]](#)
- **Pitch Shifting:** To change the pitch without altering duration, a common Phase Vocoder strategy involves a two-step process: first, time-stretch the signal by a factor corresponding to the desired pitch shift, and then resample the stretched signal (which inherently changes both pitch and duration) back to its original length. For instance, to double the perceived pitch, the sound might be time-stretched to twice its original length, and then played back at double speed.

### Basic Principles Behind the Phase Vocoder Algorithm:

The Phase Vocoder algorithm typically works through three main stages:

#### 1. Analysis:

- The incoming audio is divided into short, overlapping segments using a window function.
- An FFT is performed on each segment. This yields the magnitude and phase for various frequency components within that short time window. Repeating this for all segments gives the STFT of the signal. [\[3\]](#)

## 2. Modification:

- This stage is central to the Phase Vocoder's effectiveness. If one were to simply replay the STFT frames (the spectral snapshots) at a different rate for time stretching, it would cause phase mismatches, leading to "phasiness" or metallic sounds.
- To avoid this, the algorithm estimates the **instantaneous frequency** for each frequency component. This is essentially how fast the phase of that component is changing from one analysis frame to the next. [4]
- When creating the output signal with a new timing (i.e., a different synthesis hop size), the phase for each frequency component in a new synthesis frame is not just taken from an analysis frame directly. Instead, it's calculated by taking the phase from the *previous synthesis frame* and advancing it based on the measured instantaneous frequency and the duration of the *new synthesis hop*. This careful propagation of phase ensures that the individual frequency components in the reconstructed signal connect smoothly, preserving the sound's naturalness as much as possible. [5]
- Mathematically, if  $\phi_s(t - R_s)$  was the phase in the prior synthesis frame and  $\omega_i$  is the instantaneous frequency from the analysis stage, the new phase  $\phi_s(t)$  is updated by adding  $\omega_i \cdot R_s$ , where  $R_s$  is the synthesis hop size. The magnitudes of the frequencies are typically taken from the original analysis frames or interpolated.

## 3. Synthesis:

- After the spectral information (magnitudes and newly calculated phases) for each frame is determined, an Inverse FFT (IFFT) is applied to each frame. This converts the modified frequency-domain data back into a time-domain signal segment.
- Finally, these synthesized time-domain segments are combined, usually through an overlap-add procedure, to form the complete, modified output audio signal. [3]

More advanced Phase Vocoder implementations often include techniques like "phase locking," which aim to maintain the phase relationships between different frequency bins that belong to the same harmonic component of a sound. This can further improve the perceived quality of the processed audio, particularly for musical tones. [5]

### Question 5

Discuss the limitations of the Phase Vocoder and possible artifacts that can occur during time stretching or pitch shifting.

**Ans:**

The phase vocoder (PV) is a widely spread technique for processing audio signals. It employs a short-time Fourier transform (STFT) analysis-modify-synthesis loop and is typically used for time-scaling of signals by means of using different time steps for STFT analysis and synthesis. the explanation of the **limitations of the Phase Vocoder** and the **artifacts** that can happen during **time stretching or pitch shifting**.

### Limitations of the Phase Vocoder

**1. Phase Coherence Issues:** It analysis the sound in a thick, solid piece of something and tracking the phases of different frequencies, But it doesn't keep the phases of related frequencies in sync, then the result can be a sound blurry we can notice with quick sounds like drum hits or speech consonants.

**2. Poor Transient Handling:** The algorithm assumes the sound changes gradually, which works fine for smooth or steady sounds. But it suddenly struggles when the sharp sounds hit or a "t" sound in speech. These transients can lose their crispness and end up sounding mushy or flattened.

**3. Time-Frequency Resolution Tradeoff:** The phase Vocoder is usead as the **Short-Time Fourier Transform (STFT)**, which breaks the sound into time windows. The time windows are two types they are:

- **Long windows** give better frequency details but can't capture quick changes in time.
  - **Short windows** give better timing precision because they analyze the sound over small time slices so they can accurately detect when drum hits. But the tradeoff is that these short slices don't contain enough frequency information details to clearly distinguish between closely spaced pitches.

- **4. Aliasing and Spectral Leakage:** When you break up a signal then you need to apply a window to each segment. If it doesn't done then carefully use the right window type or enough overlap, you can get unwanted effects like **aliasing** or **spectral leakage**
- **5. Limited Quality in Pitch Shifting:** Pitch shift audio uses the phase vocoder, then it changes the pitch by altering frequency components of the sound, because it doesn't distinguish between pitch and formants. This can make someone's voice sound cartoonish or robotic. More advanced techniques, like PSOLA or wavelet-based methods, are better at preserving those natural vocal characteristics.
- **Artifacts You Might Hear During Time Stretching or Pitch Shifting**
  - **1. Phasiness or Smearing:** where the sound appears blurred. This happens because of the phase vocoder changes the timing and frequency of the audio and during the process, so it can lose the phase relationships between harmonics or partials.
  - **2. Transient Degradation** You might notice that sharp, snappy sounds (like claps or drum hits) don't hit as hard. That's because the stretching spreads these fast events out in time, dulling their impact.
  - **3. Robotic or Metallic Sound:** In Vocals, the pitch is shifted one place another place then the result can be sound artificial (robot or metallic sound) This happens because the natural harmonic relationships in the sound are distorted.
  - **4. Chorus or Flanging Effects:** Sometimes, small differences in frequency across the frame can create interference patterns that sound like chorus or those swooshing effects that weren't in the original.

## References

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5. <sup>^</sup> Laroche, J., & Dolson, M. (1999). Improved phase vocoder time-scale modification of audio. *IEEE Transactions on Speech and Audio Processing*, 7(3), 323-332.  
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## ✓ Class 3: Vocoder and STFT

### Exercise 1. Task 1: Short Time Fourier Transform (STFT)

In this task, you will explore the Short Time Fourier Transform (STFT) and its application in analyzing audio signals. The objective is to gain a better understanding of the frequency content of an audio signal over time. Follow the steps below:

- Load the provided 'guitar.wav' audio file into Google Colab.
- Visualize the waveform of the audio file to get an overview of the signal.
- Apply the STFT to the audio signal using an appropriate window size and overlap. Experiment with different parameters to observe their effects.
- Display the resulting spectrogram, which represents the frequency content of the audio signal over time.
- Explore the spectrogram and analyze the changes in the frequency content of the audio signal.
- Modify the window size and overlap and observe how they affect the resolution and clarity of the spectrogram.



- Apply the inverse STFT to the modified spectrogram to obtain the reconstructed audio signal.
- Compare the reconstructed audio signal with the original audio to evaluate the quality of reconstruction. **Explain what is happening and why.**

Consider the following points during your exploration:

- Pay attention to the trade-off between time and frequency resolution when choosing the window size and overlap.
- Experiment with different window functions (e.g., Hann, Hamming) to observe their impact on the spectrogram.
- Reflect on the limitations of the STFT in capturing time-varying frequency components in non-stationary signals.



## Exercise 2. Task 2: Pitch Correction using Phase Vocoder

In this task, you will explore pitch correction using the Phase Vocoder algorithm. The objective is to correct the pitch of a singer's performance to a target frequency, such as a "perfect" 440 Hz. Follow the steps below:

- Record a short audio clip of yourself holding a constant note.
- Load the audio clip into Google Colab and visualize its waveform.
- Apply the Phase Vocoder algorithm to modify the pitch of your performance, shifting it towards the target frequency (e.g., 440 Hz).
- Adjust the pitch correction factor to achieve the desired pitch correction effect.
- Play the modified audio and compare it with the original recording to evaluate the pitch correction.
- Analyze the spectrograms of the original and modified audio to observe the changes in the frequency content and pitch accuracy.

Take note of the following considerations:

- Experiment with different pitch correction settings to find the optimal balance between naturalness and pitch accuracy.
- Reflect on the impact of pitch correction on the overall quality and perception of your performance.
- Consider the limitations and challenges of pitch correction, and discuss potential artifacts that may arise from the process.



Have fun exploring the practical application of the Phase Vocoder algorithm for pitch correction. This exercise will provide hands-on experience in modifying pitch and understanding the concepts behind pitch correction in the context of audio signal processing.